

Abstract:

Heavy metal pollution in coastal marine areas has become a growing concern due to the increase of anthropogenic activities introducing these metals into the ocean. Heavy metals like copper and lead pose a great risk to marine life because they do not biodegrade and can accumulate in organisms over time, causing disruptions to essential physiological processes like osmoregulation, behavior, and immune function. Understanding how exposure to these metals affects marine invertebrates is crucial for identifying population level effects and using them to inform conservation decisions. Here we show that copper and lead induce physiological stress in purple shore crabs (*Hemigrapsus nudus*) across osmoregulatory, behavioral, and immune responses. We found that after three weeks, metal exposed crabs had increased hemolymph osmolality and hemocyte counts, and lead exposed crabs had increased righting times. Our results suggest that heavy metals interfere with osmoregulatory balance in crabs, likely by competing with essential ions, resulting in energetic costs to the crabs as seen by increased righting times and hemocyte counts. Together, our findings signify that heavy metal exposure in intertidal crabs can cause sublethal physiological stress that could potentially reduce organism performance and result in reduced feeding, predator avoidance, and reproductive success at the population level.

Intro:

In recent years, heavy metal pollution has become of increasing concern due to an increase of urban runoff, industrial discharge, boating/ship activity, and the development of shorelines (Science Daily, 2024). Some heavy metals, like copper and lead, do not biodegrade and stay in the ocean. Though they do not usually immediately kill organisms that consume or interact with them, they can accumulate inside them over time. This can create physiological stress for the organism, affecting osmoregulation, behavior, immune function, and long-term survival. Understanding these responses is important for conservation as this type of sublethal stress can affect an organism's ability to feed, detect predators, reproduce, and have environmental resilience (Mayer-Pinto et al., 2020). If enough individuals in a population experience this over time, it could lead to reduced population health.

Crabs are especially susceptible to heavy metal pollution since they constantly interact with sediment, seawater, and prey that can all contain pollutants. They can take up and accumulate these metals in their tissues, making them good indicators of environmental contamination (Adeleke et al., 2021). The purple shore crab (*Hemigrapsus nudus*) is a common intertidal crab that inhabits the Pacific coast. Given that they live in the intertidal, they are required to maintain internal physiological balance in an environment with changing salinity, temperature, and water quality.

One way crabs do this is through osmoregulation—the maintenance of internal water and ion balance. They can pump ions in or out of their body depending on the salinity of the water they are in. However, the pathways that do this can sometimes increase the uptake of other metal ions that they do not want like lead and copper, which can then compete with essential ions or bind to areas in their gills that can then disrupt normal ion regulation. With impaired gill function, the crabs are less able to osmoregulate and may end up spending more energy to maintain their ionic balance, leaving them with less energy for other important functions and processes (Capparelli et al., 2016). Measuring hemolymph osmolality can indicate whether *H. nudus* is maintaining normal ion balance or experiencing osmoregulatory stress.

Changes in behavior and immune condition can also serve as indicators of stress in organisms like *H. nudus*. Righting time—the time it takes for a crab to flip itself over after being placed on its back—can be used as a measure of performance. A longer righting time may indicate reduced coordination, lower energy, or increased stress. The number of hemocytes present in hemolymph is another stress indicator since they have to do with immune response in crustaceans. A higher hemocyte count correlates with an active immune system while a lower value means the immune system is inactive or suppressed (Mix et al., 1980). Together, osmolality, righting time, and hemocyte count provide insight into *H. nudus*’ stress response after exposure to lead and copper.

This study aims to address the question of whether the presence of heavy metals, specifically lead and copper, affects *Hemigrapsus nudus* physiology. An increased osmolality, righting time, and hemocyte count were interpreted as indicators of elevated stress, with potential relevance to conservation and monitoring efforts for metal exposed populations.

Methods:

14 *Hemigrapsus nudus* of similar size were randomly selected for the experiment. Two crabs were used for initial hemolymph samples, and the remaining 12 crabs were placed into control, copper, and lead treatment groups, with 4 crabs per group. Approximately 29.18 grams of lead and 30.22 grams of copper were placed in their respective holding containers. All treatments were held at a salinity of 35 ppt and the same temperature. Before exposure to copper or lead, each of the crab's weight, carapace length, carapace width, righting time, and sex were recorded. 12 crabs were then tagged, and color and number of each tag was recorded. A full water change of the copper treatment group was performed every other day.

During Week 1, two crabs were randomly selected from the 14 for an initial hemolymph extraction. About 1/8 mL of hemolymph was extracted from the individuals using U-100 insulin syringes. The hemolymph samples were then transferred into centrifuge tubes, centrifuged for 10 seconds, and two drops of EDTA anticoagulant were added before being placed into the fridge to be analyzed later. The two crabs were then unused for the rest of the experiment.

During Week 2, righting time was recorded for two crabs randomly selected from each treatment group. Hemolymph was then extracted from those same crabs and centrifuged. 2 microliters of dye were added to 28 microliters of the hemolymph samples and centrifuged again. The stained samples were then added to a hemocytometer to analyze under a microscope for hemocyte counting.

During Week 3, righting time was recorded and hemolymph was extracted for all remaining crabs. Without centrifuging, all samples were observed under the microscope for hemocyte counts. All hemolymph samples from Weeks 1, 2, and 3 were then placed in the osmometer to measure hemolymph osmolality.

Results:

Osmolality:

Average hemolymph osmolality generally increased among the treatment groups from Week 2 to Week 3. The control group had a small increase from 861 mmol/kg to 924 mmol/kg, the lead group increased from 789 mmol/kg to 1017 mmol/kg, and the copper group increased from 807

mmol/kg to 982 mmol/kg. Both heavy metal treatment groups had a higher average osmolality by Week 3 compared to Week 2, with the highest increase in the lead group.

A one-way ANOVA comparing final osmolality among all treatment groups was not statistically significant ($p = 0.0633$), but a comparison between the control and lead groups was ($p = 0.0065$). This suggests that lead exposure may have had an effect on osmolality.

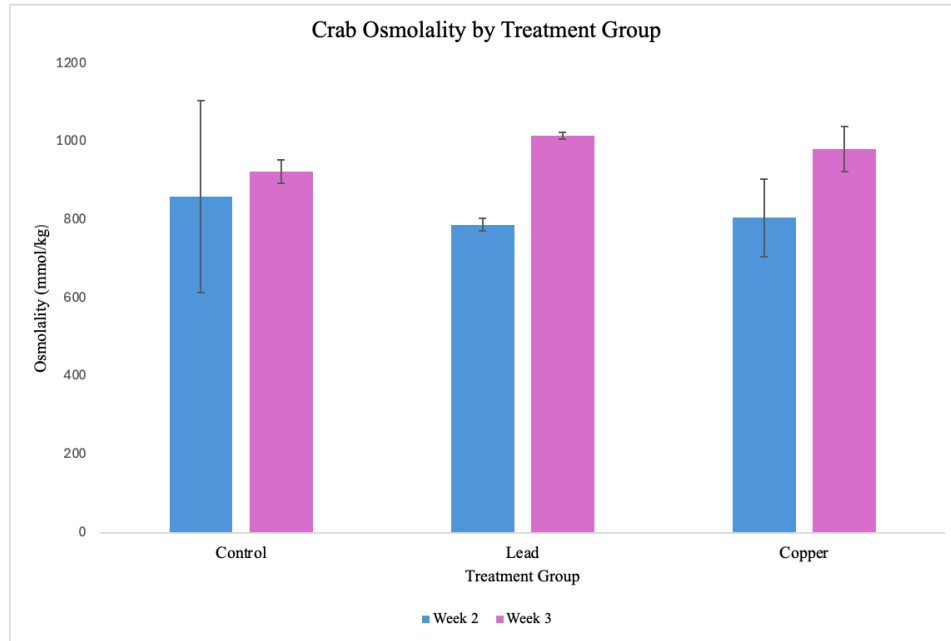


Figure 1. Bar graph showing mean hemolymph osmolality (mmol/kg) of *Hemigrapsus nudus* exposed to control, lead, and copper treatments during Weeks 2 and 3. Bars represent treatment means, and error bars represent standard deviation.

Righting time:

Average righting time differed among treatment groups and weeks. The control crabs consistently had a lower average righting time of 0.5-2.5 seconds across all weeks. The lead treatment group showed an increase in righting time from about 3 seconds in Week 1 to over 9 seconds in Weeks 2 and 3. In contrast, the copper treatment showed a decrease in righting time across weeks from about 4 seconds in Week 1 to under 1.5 seconds in Weeks 2 and 3.

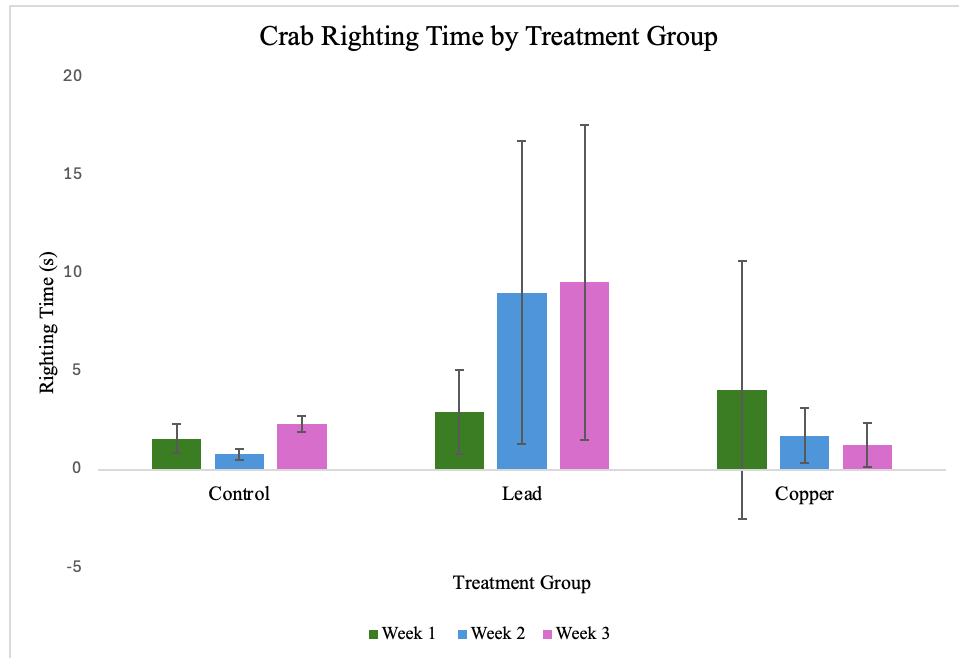


Figure 2. Bar graph showing mean righting time (s) of *Hemigrapsus nudus* exposed to control, lead, and copper treatments across three weeks. Bars represent treatment means, and error bars represent standard deviation.

Hemocyte count:

Hemocyte count differed among treatment groups. No hemocytes were observed in the control group while approximately three were in the lead group and six were in the copper group.

However, a one-way ANOVA comparing total hemocyte count among treatment groups was not statistically significant ($p = 0.3587$), so interpretation should be cautious.

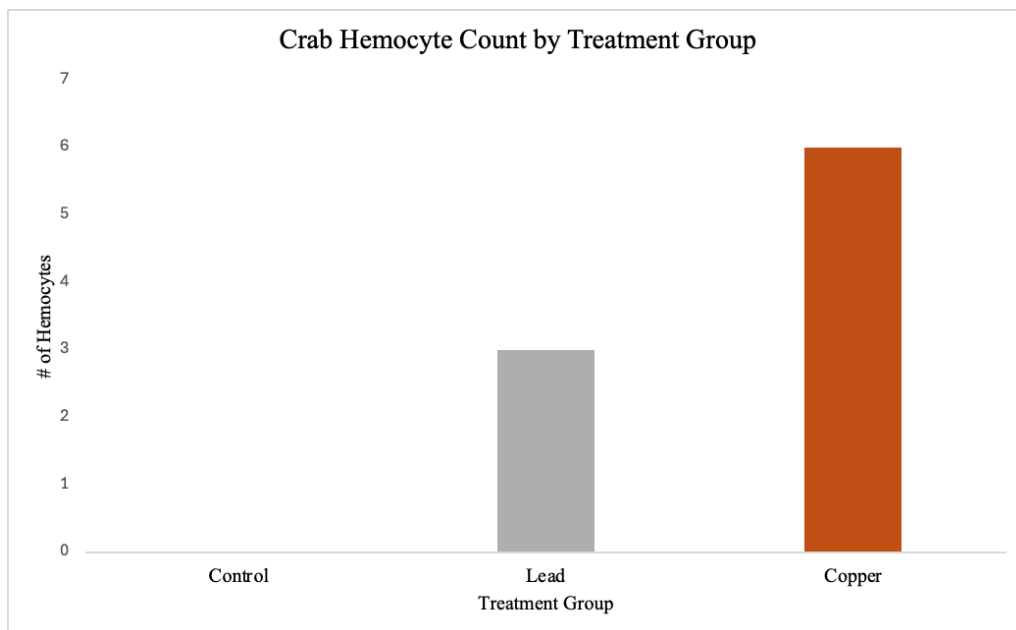


Figure 3. Bar graph showing hemocyte counts in crabs across treatment groups control, lead, and copper for Week 3 of the experiment.

Discussion:

The results of this study suggest that copper and lead exposure may cause an increase in physiological stress in *Hemigrapsus nudus*. The main findings were that heavy metal exposure may affect osmoregulation, behavior, and immune activity in crabs—all of which are stress induced responses.

Our results support previous studies that found heavy metals can cause physiological stress in crabs. Prior work found that crabs can accumulate heavy metals through osmoregulation, which can then compete with essential ions and lead to hyper-osmotic stress. This means that if the solute concentration outside of the cells is too high, water will move out of the cells causing dehydration and cell damage within the organism (Capparelli et al., 2016). Our findings that osmolality increased in the copper and lead exposed groups supports this as it is indicative of osmoregulatory stress. Although the ANOVA comparing all treatment groups was not significant, the comparison between control and lead did show a significant difference in final osmolality ($p = 0.0065$). This suggests that lead exposure may have had the strongest effect on osmoregulatory balance. The increased righting time in the lead group further suggests physiological stress since the disruption of ion balance can increase energetic costs and reduce the crab's performance. However, this was not supported by the copper group, which had a decreased righting time. Additionally, the absence of hemocytes in the control group but presence in the metal groups suggests that heavy metal exposure could have triggered immune activation in the crabs (Mix et al., 1980).

These results should, however, be interpreted with caution given several limitations to the experiment. With only 14 crabs, the sample size was considerably small, and Week 2 only sampled 2 crabs from each treatment group, making result averages very sensitive to individual variation as seen by the large standard deviations in figures. This is also supported by the statistical tests where osmolality and hemocyte count were not significantly different between all treatment groups overall, despite visible differences in the means. Additionally, methods for counting hemocytes were not consistent across weeks resulting in limited hemocyte count data. Week 1 and 2 samples were centrifuged which may have resulted in the hemocytes being destroyed while Week 3 samples were counted fresh with no centrifuging. This could have also affected osmolality values if cells were damaged in the process.

Future studies should use larger sample sizes, consistent methods when analyzing hemolymph for osmolality and hemocytes, and specific metal concentrations rather than weights of metal. A hormonal assay would also be useful to show physiological stress.

Though stronger evidence is needed, these findings are important for conservation as they suggest that heavy metal exposure—something becoming increasingly common—could be causing sublethal stress and reducing organism performance. This could lead to higher mortality rates and lower resilience in exposed populations over time.

Works Cited:

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